

SmolVLM for Dense Video Captioning

Week 9: Internship Update

Code updates: CSTA

- Optimizing: not opening video multiple times, batched feature extraction
- Including presampling and postsampling (can take max_frames and max_input_frames as an input)
- Works with up to 15% of the most 'important' frames by length, or with those whose score is higher than x%

Paper updates: Related work

- Small VLMs in general:
 - Architectural blocks: Visual Encoder, LLM, Connector
 - Data preprocessing, tokenization techniques

Paper updates: Related work

- Ovis2:
 - Uses linear mapping for probabilistic visual tokens and an additional visual embedding look-up table, instead of connector for projecting visual embeddings into the same vector space as textual embeddings

$$\mathbf{v}_i = \text{softmax}(\mathbf{W} \mathbf{r}_i), \quad \mathbf{W} \in \mathbb{R}^{K \times d}$$

- K is visual vocabulary size
- Probability distribution over K visual words
- This captures how related $\mathbf{r}_i$ is to each of words from visual vocabulary

Paper updates: Related work

- Ovis2:
 - Visual embedding table
 - Each visual word word is associated with an embedding vector $\mathbf{e}_k$
 - Each visual token is embedded based on probabilistic token:

$$\mathbf{V}_i = \sum_{k=1}^K v_{i,k} \mathbf{e}_k \in \mathbb{R}^{d'}$$

$$\mathbf{V}_i = \mathbb{E}_{k \sim \mathbf{v}_i}[\mathbf{e}_k].$$

- Three training stages, freezing different parts of the architecture each time

Paper updates: Related work

- Qwen-VL:
 - Only the ability of processing images and text
 - Based on Qwen LLM
 - Position-aware vision-language adapter that is based on cross-attention

Paper updates: Related work

- Qwen2-VL:
 - Three sizes: Qwen2-VL-2B, Qwen2-VL-7B and Qwen2-VL-72B
 - LLM: Qwen2
 - Naive dynamic resolution support enables Qwen2-VL to process images of any resolution, dynamically converting them into a variable number of visual tokens
 - 2D Rotary Position Embedding
 - Multimodal Rotary Position Embedding (M-RoPE): temporal, height and width
 - For textual input, these components are identical position IDs, for images, the temporal component remains the same and for videos, the temporal component increases with every new frame
 - Two frames per second

Paper updates: Related work

- Qwen2.5-VL:
 - Use cases: faster computation
 - Window attention in ViT
 - Converts images of varying sizes into token sequences of corresponding lengths, using actual image dimensions rather than normalized coordinates
 - Dynamic frame sampling for videos
 - Multimodal Rotary Position Embedding Aligned to Absolute Time

Paper updates: Related work

- SmolVLM:
 - SmolVLM-2.2B, SmolVLM500M and SmolVLM-256M
 - Tokenization strategies and data curation
 - Multimodal models benefit from a balanced encoder-LM parameter allocation
 - Self-attention mechanism which visual tokens from the vision encoder are concatenated with textual tokens and jointly processed by a language mode
 - Pixel shuffling
 - Image splitting

Benchmark testing

- VideoMME